

CYCLISTIC BIKE-SHARE

Case Study 1: How does a bike-share navigate speedy success?

2019 SQL Analysis & Marketing Insights

Full-year 2019 | MySQL analysis | Excel visual reporting

Prepared as a portfolio capstone case study

Executive Summary

Business question. The case study asks how annual Members and Casual riders use Cyclistic bikes differently. The case-study brief assigns this first question to the analyst and expects the findings to support marketing strategy. The source brief also states that annual members are more profitable than casual riders and that the marketing objective is to explore converting casual riders into annual members. (Case Study brief, pp. 2–3.)

Metric	2019 result	Interpretation
Total trips	3,818,004	Full-year 2019 trip population
Member trips	2,937,367 (76.9%)	Members generated most recorded rides
Casual trips	880,637 (23.1%)	A substantial non-member segment
Average ride length — Member	12.93 min	Shorter average rides
Average ride length — Casual	39.43 min	About 3.05× the member average
Peak hour	17:00 — 475,154 rides	Highest overall hourly demand
Peak month	August — 590,184 rides	Highest monthly demand
Peak quarter	Q3 — 1,640,718 rides	Highest quarterly demand

Core story. The 2019 data shows a clear behavioral difference between the two user groups: Members account for most rides, while Casual riders take substantially longer rides and a larger proportion of their rides on weekends. Casual usage also rises sharply during the warmer months. These differences identify useful moments for membership messaging, but the trip data alone cannot establish why a casual rider would purchase a membership or whether a particular digital campaign causes conversion.

1. Ask — Business Task

Business task: Understand how annual Members and Casual riders use Cyclistic bikes differently, using trip behavior to identify evidence-based opportunities for membership-conversion marketing.

Stakeholders: Cyclistic marketing leadership, the marketing analytics team, and the executive team approving the proposed marketing program. The case study brief identifies the marketing director as the project manager and the executive team as the approval audience.

2. Prepare — Data & Scope

Item	Project treatment
Primary source	Cyclistic 2019 Q1, Q2, Q3 and Q4 trip CSV files provided for the project.
Scope	Full-year 2019; 3,818,004 trips.
Primary analysis tool	MySQL / MySQL Workbench.
Reporting tool	Excel workbook with summary tables and charts.
Station reference	Station IDs retained in trip analysis; station names used for station summaries were taken from 2019 trip records.
Privacy	No personally identifiable information was used.

Source note. The case-study brief describes the historical trip data as public data made available by Motivate International Inc. and notes that privacy restrictions prevent use of personally identifiable information. (Case Study brief, p. 3.)

3. Process — Cleaning & Transformation

- Imported the four 2019 quarterly CSV files into a common MySQL staging table.

- Added a period field so each trip can be traced to 2019 Q1, Q2, Q3 or Q4.
- Converted start and end timestamps into proper DATETIME values.
- Converted trip duration from seconds to minutes for the ride-length analysis.
- Retained all 3,818,004 raw trips. The 1,848 trips longer than 24 hours (0.05%) were retained in the raw data but assigned NULL for the cleaned ride-length metric so they would not distort duration statistics.
- Blank or invalid birth-year values were treated as NULL for age analysis; the associated trips were retained.
- Standardized user types for reporting: Subscriber → Member; Customer → Casual.
- Checked quarter-level trip IDs for duplicates; none were found.
- Core fields contained no SQL NULL values in the direct completeness checks; blank demographic strings were handled separately.

4. Analyze — Findings

4.1 Member vs Casual behavior

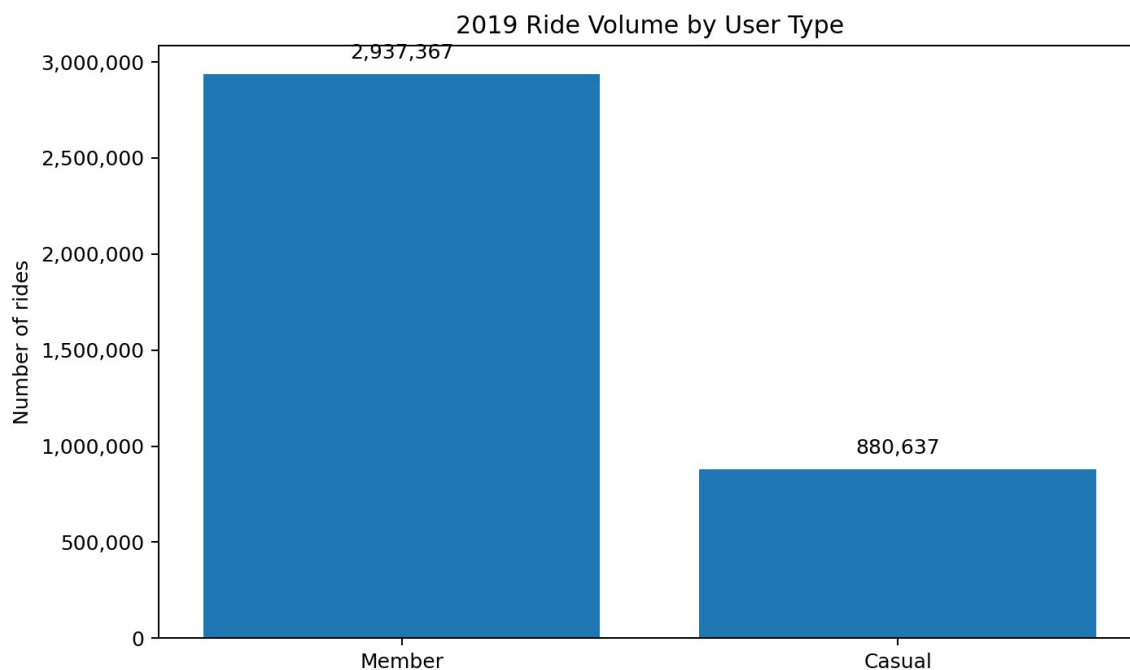


Figure 1. Total 2019 rides by user type

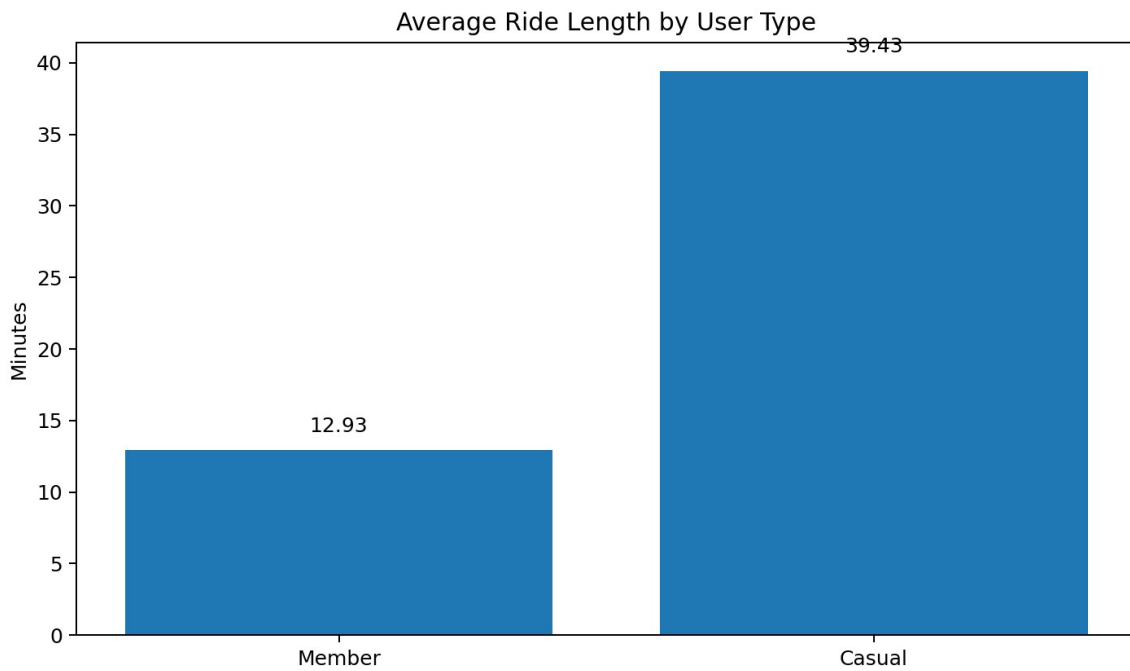


Figure 2. Average ride length by user type

Finding. Members made up 76.9% of all recorded rides, while Casual riders accounted for 23.1%. The average Casual ride was 39.43 minutes versus 12.93 minutes for Members, a difference of about 3.05 $\times$. The data therefore shows a strong behavioral distinction in ride duration between the groups.

4.2 Day-of-week and weekend behavior

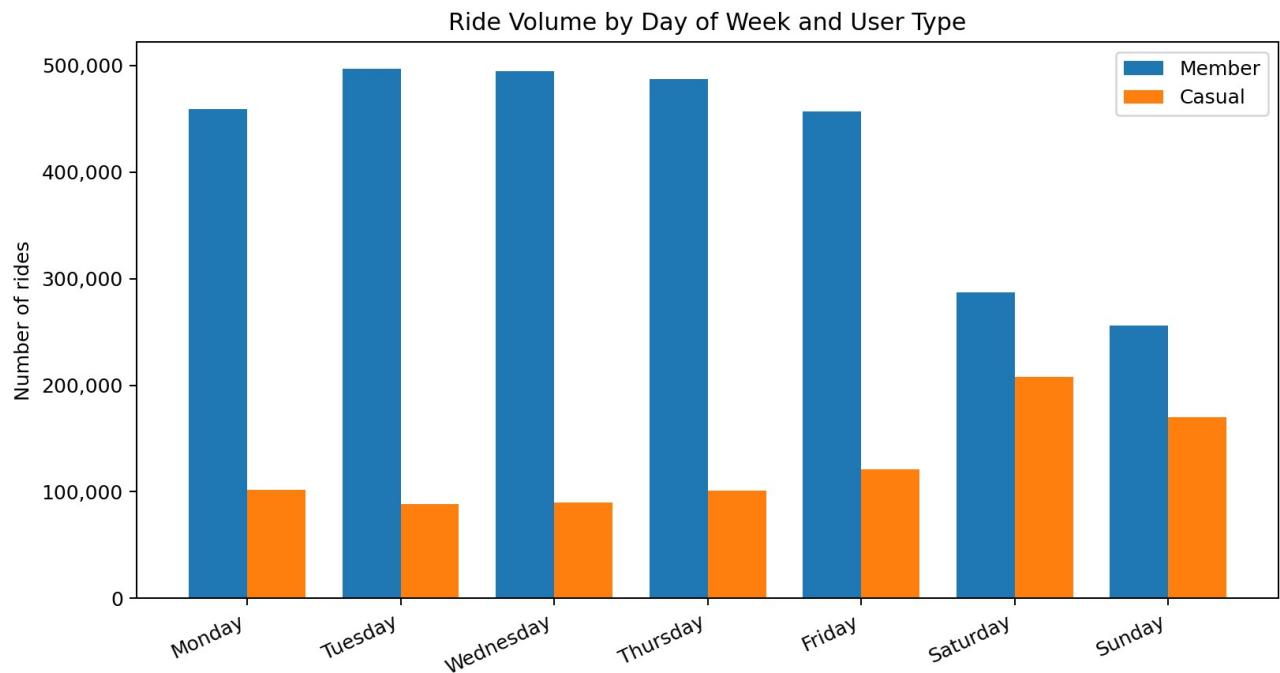


Figure 3. Ride volume by day of week and user type

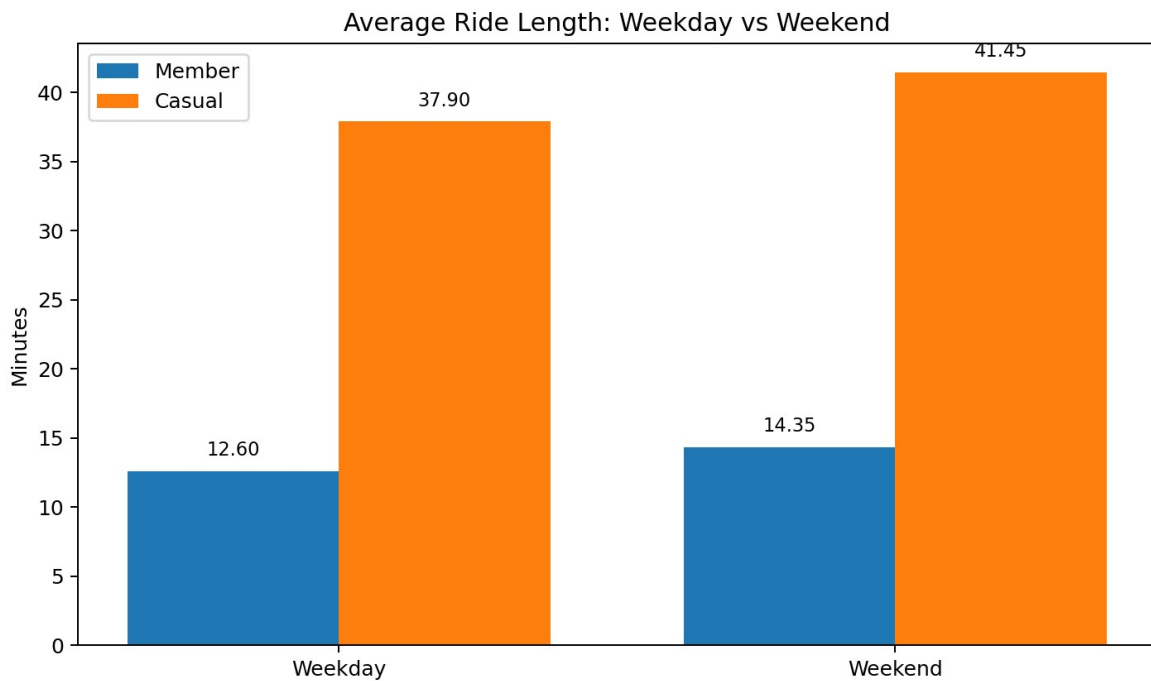


Figure 4. Average ride length on weekdays vs weekends

Finding. Casual riders show a much stronger weekend presence than Members. About 42.9% of Casual rides occurred on weekends, compared with 18.5% of Member rides. Average ride length also rises on weekends for both groups: 37.90 to 41.45 minutes for Casual riders and 12.60 to 14.35 minutes for Members.

4.3 Monthly and quarterly patterns

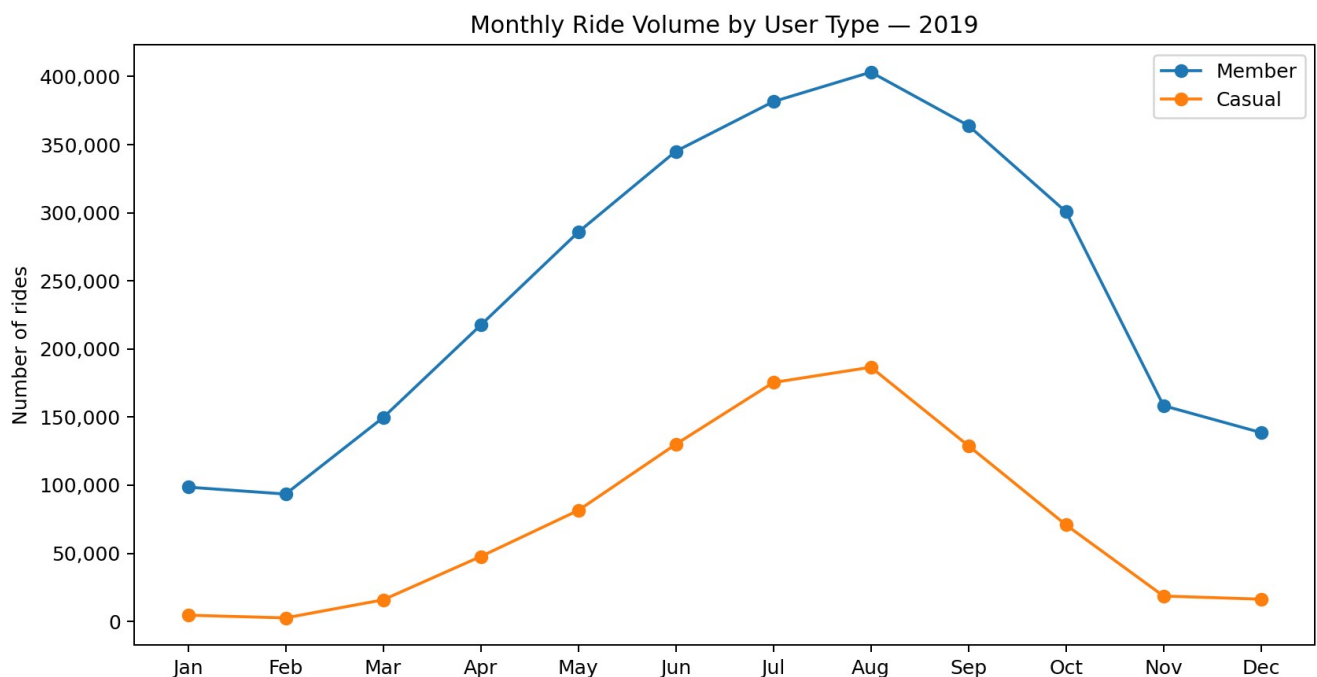


Figure 5. Monthly ride volume by user type — 2019

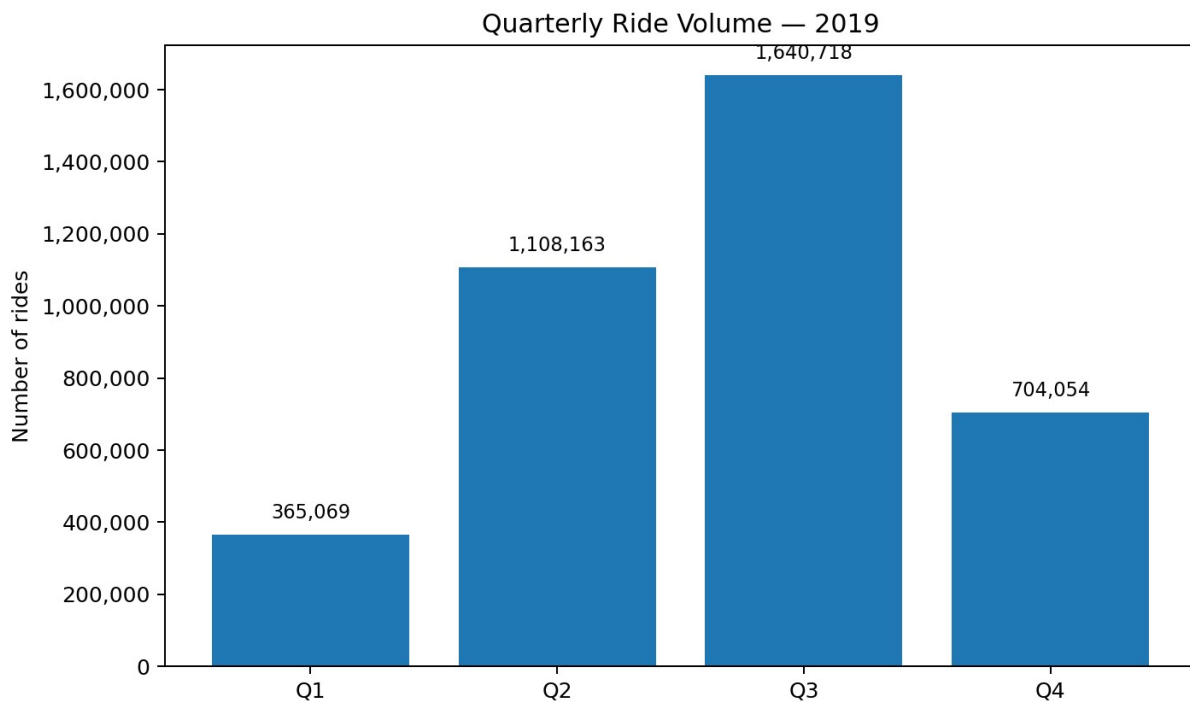


Figure 6. Quarterly ride volume — 2019

Finding. Demand is strongly seasonal. Total rides rise from winter into the summer period, peak at 590,184 in August, and then decline. Q3 is the highest-volume quarter at 1,640,718 rides. Casual usage rises especially sharply through the warmer months, while Member usage remains materially higher throughout the year.

4.4 Hourly demand

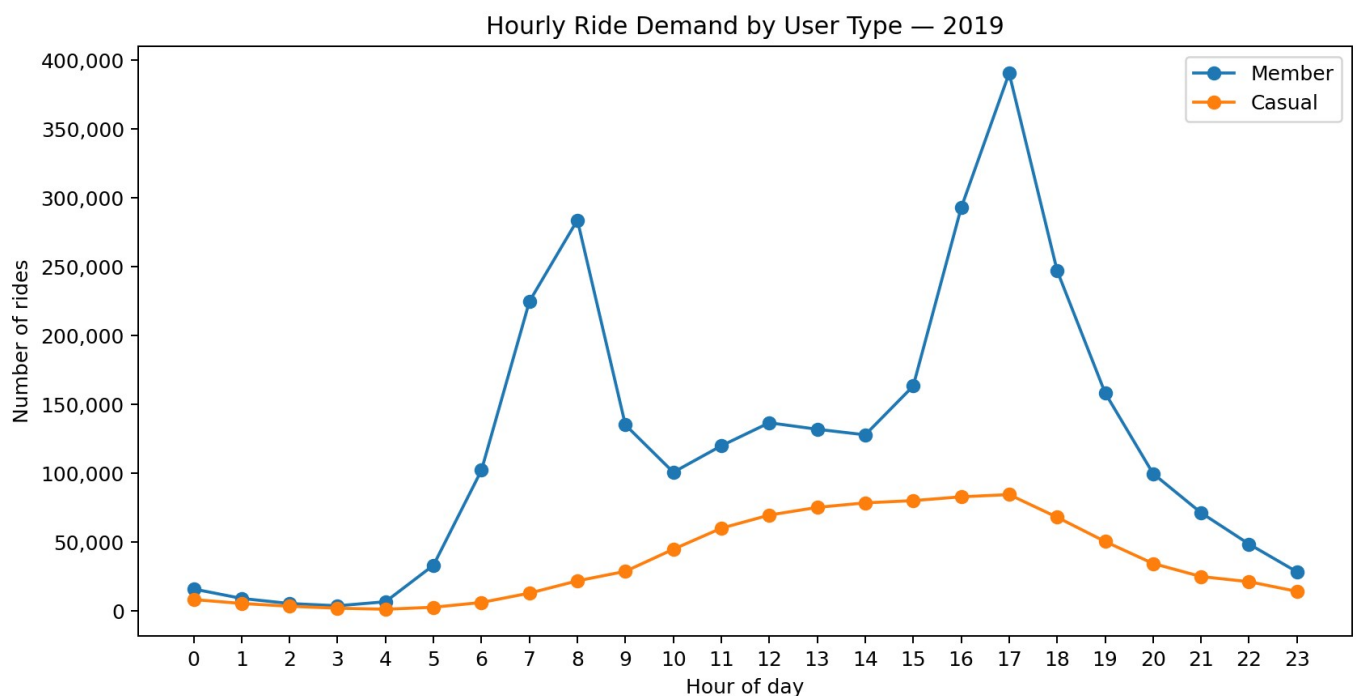


Figure 7. Hourly ride demand by user type — 2019

Finding. The overall peak is 17:00 with 475,154 rides. Both Members and Casual riders also peak at 17:00 in the user-type analysis. Member demand is especially strong during the weekday commute-oriented hours, while Casual usage is more distributed into the afternoon and evening.

4.5 Ride-length distribution

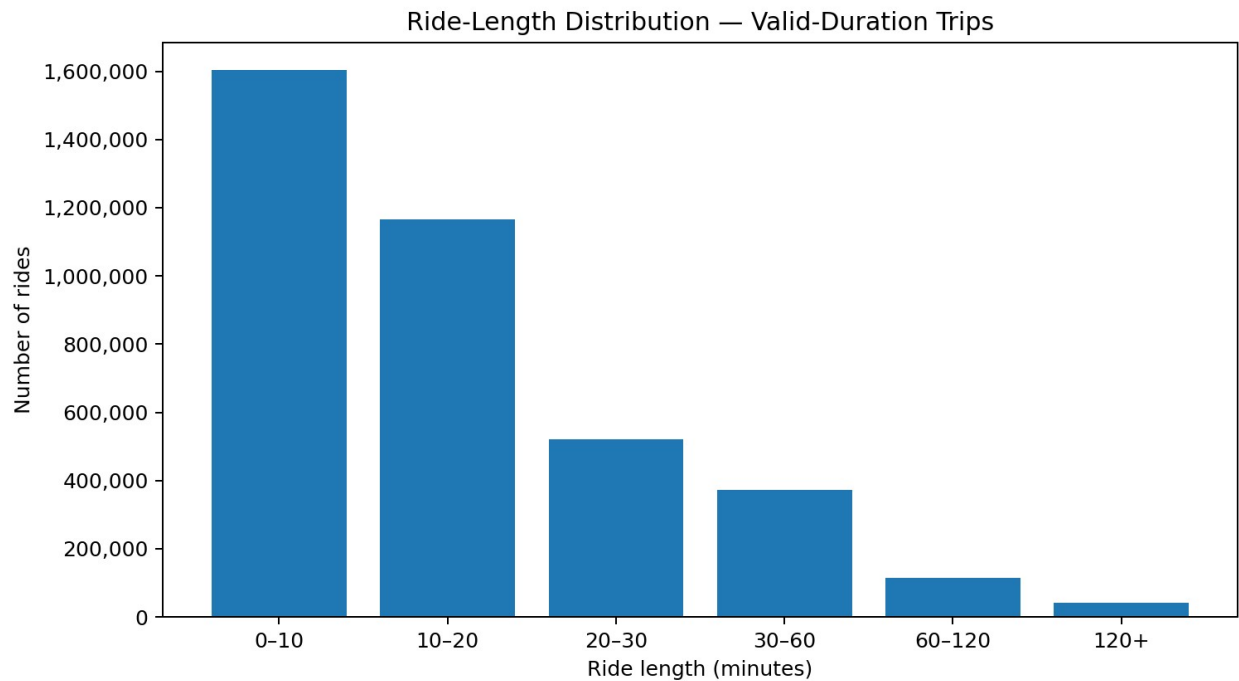


Figure 8. Distribution of valid-duration rides

Finding. Among the 3,816,156 rides with valid ride length, 42.0% lasted under 10 minutes and 72.6% lasted under 20 minutes. Overall, 86.2% lasted under 30 minutes. The average ride length of 19.03 minutes is therefore supported by a distribution concentrated in shorter rides, with a smaller long-ride tail.

4.6 Age and gender — supporting analysis

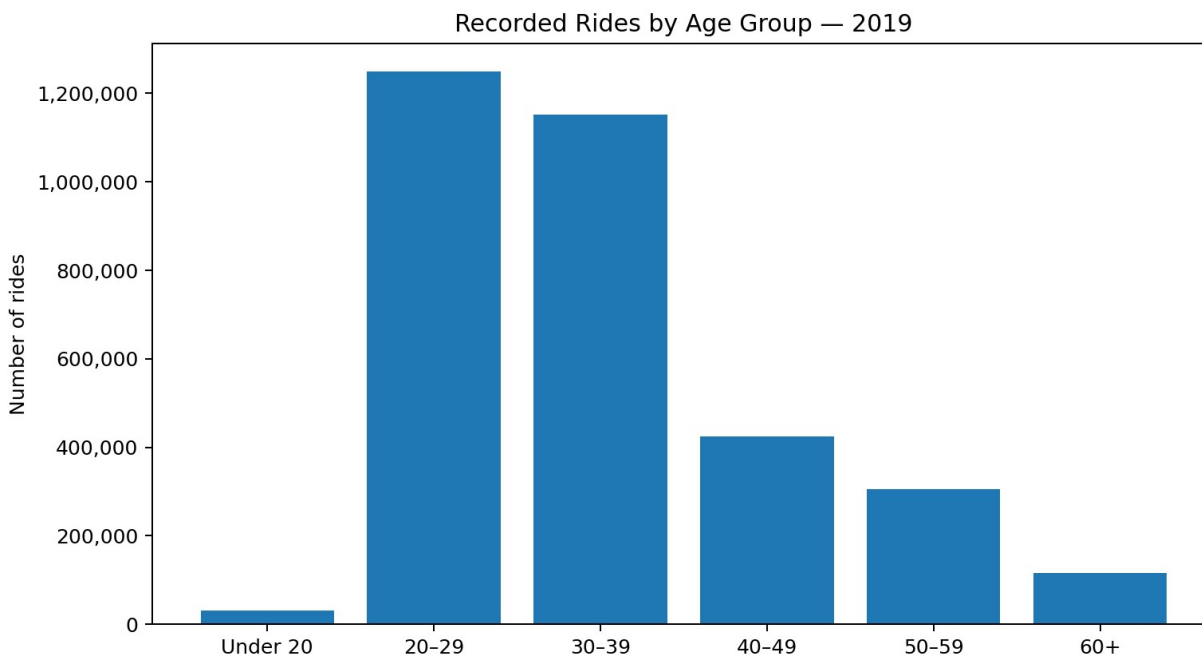


Figure 9. Recorded rides by age group; Unknown excluded from the chart

Age. Recorded rides are concentrated in the 20–39 age groups. Age analysis excludes unknown/invalid birth years from age-specific interpretation, while the underlying trips remain in the dataset.

Gender. Gender is incomplete in the source: 559,206 trips (14.6%) have no recorded gender. Among recorded-gender trips, the Male category has the largest ride count, followed by Female. This should be treated as a descriptive pattern rather than evidence of a causal demographic effect.

5. Supporting Station & Route Analysis

Station analysis was treated as supporting context rather than the main business question. The 2019 trip records were used to rank high-volume station origins, destinations, and routes.

Rank	Top starting station	Rides
1	Streeter Dr & Grand Ave	67,983
2	Canal St & Adams St	54,389
3	Clinton St & Madison St	49,908
4	Lake Shore Dr & Monroe St	49,804
5	Clinton St & Washington Blvd	48,153
6	Columbus Dr & Randolph St	39,192
7	Michigan Ave & Washington St	37,696
8	Michigan Ave & Oak St	35,449
9	Lake Shore Dr & North Blvd	34,472
10	Franklin St & Monroe St	34,297
Rank	Top ending station	Rides
1	Streeter Dr & Grand Ave	81,723
2	Clinton St & Washington Blvd	50,686
3	Canal St & Adams St	50,127
4	Clinton St & Madison St	47,311
5	Lake Shore Dr & North Blvd	42,459

6	Lake Shore Dr & Monroe St	40,633
7	Michigan Ave & Washington St	40,331
8	Michigan Ave & Oak St	37,849
9	Millennium Park	37,424
10	Theater on the Lake	35,939
Rank	Route	Rides
1	Lake Shore Dr & Monroe St → Streeter Dr & Grand Ave	9,872
2	Shedd Aquarium → Streeter Dr & Grand Ave	3,657
3	Canal St & Adams St → Michigan Ave & Washington St	3,649
4	Streeter Dr & Grand Ave → Millennium Park	3,376
5	Streeter Dr & Grand Ave → Lake Shore Dr & Monroe St	3,319
6	Streeter Dr & Grand Ave → Theater on the Lake	3,317
7	Streeter Dr & Grand Ave → Lake Shore Dr & North Blvd	3,081
8	Millennium Park → Streeter Dr & Grand Ave	3,077
9	Theater on the Lake → Streeter Dr & Grand Ave	2,840
10	Michigan Ave & Washington St → Clinton St & Washington Blvd	2,840

Station-network note. The analysis identified 152,247 same-station trips (3.99%) and 3,665,757 different-station trips (96.01%). Same-station records were excluded from the “top routes” definition so that routes represent movement between two different stations.

6. Act — Three Recommendations

1. Focus membership-conversion tests on weekends and the warm-season peak. Casual riders account for 42.9% of their rides on weekends, versus 18.5% for Members, and Casual rides are longer on weekends (41.45 minutes). Casual volume also peaks in the summer period. Test membership messaging and offers during these high-engagement periods.

2. Use high-traffic locations and the late-afternoon demand window for targeted messaging. 17:00 is the overall peak hour at 475,154 rides, and several of the highest-volume stations are recurring trip endpoints/origins. Use high-traffic locations and the late-afternoon window as test environments for membership prompts, then compare conversion rates across locations and messages.

3. Build and measure a leisure-oriented digital campaign for Casual riders. Casual rides are materially longer than Member rides and show stronger weekend and seasonal variation. Test digital creative that reflects leisure-oriented use during the May–August high-volume period, with campaign exposure and membership sign-up tracking so the effect can be evaluated rather than assumed.

7. Limitations & Next Data Needed

- The analysis answers the assigned question about behavioral differences, but the trip data does not contain membership purchase history, price-plan details, campaign exposure, or conversion outcomes.

Therefore it cannot establish why a specific casual rider would buy a membership or which digital tactic will convert them.

- 2019 is the analysis scope used for this SQL capstone implementation. Broader multi-year analysis could reveal whether the patterns persist over time.
- Age and gender fields contain missing/unknown values and are therefore supporting rather than primary evidence.
- Ride durations over 24 hours were treated as outliers for duration metrics. Keeping them in raw data preserves traceability while preventing extreme values from distorting averages.
- For stronger marketing recommendations, add membership pricing, purchase/conversion data, campaign-touch data, and ideally survey data on rider motivations.

8. Conclusion

Conclusion. The analysis answers the case study’s first business question: Members and Casual riders use Cyclistic differently. Members generate most rides and have much shorter average rides, while Casual riders take longer rides and exhibit a stronger weekend and seasonal pattern. The clearest marketing opportunity supported by the data is therefore to test membership-conversion messaging at the moments when Casual usage is highest—especially weekends, warm-season months, and high-demand time windows—while measuring actual conversion before scaling any tactic.

Appendix — Key 2019 Results

Area	Result
Total trips	3,818,004
Mean ride length	19.03 minutes
Maximum analytical ride length	1,440 minutes
Trips >24 hours	1,848 (0.05%)
Mode day	Thursday — 588,287 rides
Peak hour	17:00 — 475,154 rides
Peak month	August — 590,184 rides
Peak quarter	Q3 — 1,640,718 rides
Casual avg ride length	39.43 minutes
Member avg ride length	12.93 minutes
Weekend Casual average	41.45 minutes
Weekend Member average	14.35 minutes
Valid-duration rides	3,816,156
Same-station trips	152,247 (3.99%)
Different-station trips	3,665,757 (96.01%)

Reference: Cyclistic Case Study 1: “How does a bike-share navigate speedy success?” course brief, especially pages 2–8 (Ask, Prepare, Process, Analyze, Share, Act).